

MUHAMMAD AKHTAR

Senior Software Engineer | AI, Python & Backend Architecture

akhtarm821@gmail.com | +92 309 8638327 | [linkedin.com/in/muhammad-akhtar-47b1a4102](https://www.linkedin.com/in/muhammad-akhtar-47b1a4102) | Multan, Pakistan

PROFESSIONAL SUMMARY

Full Stack & AI Software Engineer with 6+ years of experience designing scalable backend systems, AI-driven applications, and automation platforms using Python (Django, FastAPI) and PHP (Laravel). Skilled in integrating Large Language Models (LLMs), Computer Vision, and Agentic AI workflows to automate complex enterprise processes and enhance product capabilities.

- **Generative & Agentic AI:** Architected AI-driven donor management and enrichment platforms delivering real-time analytics and 80% enrichment accuracy.
- **Computer Vision:** Engineered automated image segmentation and dynamic layer editing systems to streamline customized graphic workflows.
- **System Architecture:** Designed and built a Django-based University Strategy Management System end-to-end, tracking 150+ objectives and 200+ KPIs with multi-level approval workflows.
- **Performance Optimization:** Reduced API response time from 14 seconds to 2 seconds through Elasticsearch integration and backend refactoring.

TECHNICAL SKILLS

- **AI & Machine Learning:** Generative AI, Large Language Models (LLMs), OpenAI API, Agentic AI Workflows, Computer Vision, Image Segmentation, Prompt Engineering
- **Languages & Frameworks:** Python, Django, Django REST Framework, FastAPI, PHP, Laravel, Codeigniter, Lumen
- **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, Redis
- **Frontend & Mobile:** JavaScript, React, Next.js, jQuery, HTML, CSS, Bootstrap, Tailwind CSS, React Native, Expo
- **Automation & DevOps:** REST API Design, 3rd-Party API Integration, N8N, Chrome Extensions (Manifest V3), AWS EC2, GitHub Actions, CI/CD
- **Security & Auth:** RBAC, OAuth 2.0, Authentication, Authorization

PROFESSIONAL EXPERIENCE

Vaival Technologies LLC

March 2022 – June 2026

Lead Software Engineer – AI, Backend Architecture & Automation

AI-Powered Image Segmentation & Layer Editing System (Computer Vision)

- Developed a computer vision-based image processing solution to automatically separate images into background, foreground, and text components.
- Implemented segmentation and masking techniques to enable independent editing of different image regions.
- Enabled dynamic color transformation of selected layers while preserving text quality and visual consistency.
- Improved image customization workflows by automating complex manual editing tasks.

AI-Powered Donor Data Enrichment Platform (Agentic AI, FastAPI, PostgreSQL)

- Architected an AI-driven donor enrichment platform using FastAPI and PostgreSQL supporting multi-campaign operations and agentic workflows.
- Built a multi-agent enrichment engine integrating RocketReach & TruthFinder APIs, achieving 80% enrichment success and reducing research effort by 30+ hrs/week.
- Designed scalable REST APIs with a Next.js frontend, lowering API latency by 20%.

- Created a Manifest V3 Chrome extension (RaizelQ) with Redis-backed distributed job queues, processing 200+ tasks daily; lowered manual intervention by 90%.

Generative AI Email Assistant & Sendpad SaaS (OpenAI API, Laravel, React)

- Integrated OpenAI LLMs into applications to generate personalized email drafts based on user requirements and contextual inputs.
- Implemented prompt-based generation to improve content quality, consistency, and creation speed.
- Built Sendpad email marketing SaaS with Stripe subscriptions and AI generation, driving 90 paid subscriptions in Q1.

University Strategy Management System (Django, PostgreSQL)

- Architected and built an end-to-end strategy management platform tracking 150+ objectives and 200+ KPIs across university departments.
- Implemented sequential multi-level approval workflows with role-based access control (RBAC), Django templates, and Excel reporting export with date filters.

Data Integration & HR Workflow Automation (FastAPI, AWS, N8N, Supabase)

- Deployed a FastAPI platform on AWS EC2 consolidating 7+ data sources (11GB data) into MongoDB and BigQuery with zero-downtime CI/CD.
- Automated employee synchronization across HR & payroll systems via N8N and webhooks, accelerating provisioning by 95% and saving \$30K annually.

Additional Projects (eCommerce & Driver Apps)

- **Cartzy APIs:** Optimized Laravel REST APIs with Elasticsearch, cutting product search time from 14s to 2s.
- **WCL Driver Pro:** Created React Native app for 20+ drivers handling 250+ daily deliveries with barcode scanning.

Bitsoftsol

March 2020 – December 2022

Full Stack Developer (PHP/Laravel)

- Delivered full-stack HR & workflow features including 15-language multilingual support and Lumen desktop-sync (reduced latency by 30%).
- Built bulk S3 image-processing APIs handling 6,000+ daily images.

Graymatrix

January 2020 – December 2021

Full Stack Developer (PHP/Laravel)

- Integrated BigBlueButton for virtual classrooms and Zoho CRM via OAuth 2.0 for secure bidirectional synchronization.
- Rewrote raw MySQL queries using PDO to prevent SQL injection and conducted R&D on Zoho REST APIs and PayPal gateways.

EDUCATION

Bachelor of Science in Computer Science

Institute of Southern Punjab, Multan (2015 – 2019)

LANGUAGES

English, Urdu